Advanced Mathematical Techniques
Problem Set 6

1.
$$\int_0^{\frac{\pi}{2}} \sin^4\theta\cos^6\theta d\theta = \frac{\Gamma(\frac{5}{2})\Gamma(\frac{7}{2})}{2\Gamma(6)} = \boxed{\frac{3\pi}{512}}$$

2.
$$\int_0^{\frac{\pi}{2}} \sin^5\theta\cos^6\theta d\theta = \frac{\Gamma(3)\Gamma(\frac{7}{2})}{2\Gamma(\frac{13}{2})} = \boxed{\frac{8}{693}}$$

3.
$$\int_0^{\frac{\pi}{2}} \frac{\ln\sin\theta}{\sin\theta} d\theta$$
$\boxed{\text{diverges}}$ at 0 because of the natural log's weakness, since $\sin\theta \approx \theta$ near 0.

4.
$$t = \sin\theta \implies \int_0^{\pi/2} \frac{\ln\sin\theta}{\cos\theta}\,d\theta = \int_0^1 \frac{\ln t}{1-t^2}\,dt = \sum_{n=0}^{\infty} \underbrace{\int_0^1 t^{2n}\ln t\,dt}_{-1/(2n+1)^2} = -\sum_{n=0}^{\infty}\frac{1}{(2n+1)^2} = -\lambda(2) = \boxed{-\frac{\pi^2}{8}}$$

5.
$$\int_0^1 \frac{\ln^2(1-t)}{t\sqrt{t}}dt \to \int_0^1 t^{-\frac{3}{2}}(1-t)^{\varepsilon} = \frac{\Gamma(-\frac{1}{2})\Gamma(1+\varepsilon)}{\Gamma(\frac{1}{2}+\varepsilon)} = \frac{(\delta-\frac{1}{2})^{-1}\Gamma(1+\varepsilon)\Gamma(\frac{1}{2}+\delta)}{\Gamma(\frac{1}{2}+\varepsilon+\delta)} = \frac{(\delta-\frac{1}{2})^{-1}\Gamma(1+\varepsilon)\Gamma(\frac{1}{2}+\delta)}{2^{-2\varepsilon-2\delta}}$$
$$= \frac{(\delta-\frac{1}{2})^{-1}}{2^{2\delta}}\sqrt{\pi}\frac{\Gamma(1+2\delta)}{\Gamma(1+\delta)}\frac{\Gamma(1+\varepsilon)2^{2\varepsilon}2^{2\delta}\Gamma(1+\varepsilon+\delta)}{\sqrt{\pi}\Gamma(1+2\varepsilon+2\delta)}$$
$$= -2(1+2\delta+4\delta^2+\cdots)(1+2\varepsilon\ln 2+2\ln^2 2\varepsilon^2+\cdots)\exp\left(\sum_{k=2}^{\infty}\frac{(-1)^k\zeta(k)}{k}\left[(2^k-1)\delta^k+\varepsilon^k-(2^k-1)(\varepsilon+\delta)^k\right]\right)$$
$$\xrightarrow{\text{taking the 2nd power coeff}} 2\left(-4\ln^2 2+\frac{\pi^2}{6}\cdot\frac{1}{2}\right) = \boxed{\frac{2\pi^2}{3}-8\ln^2 2}$$

6.
$$\int_0^1 \frac{\ln t\ln(1-t)}{t}dt = -\sum_{n=1}^{\infty}\frac{1}{n}\int_0^1 t^{n-1}\ln t dt = \sum_{n=1}^{\infty}\frac{1}{n^3} = \boxed{\zeta(3)}$$

7.
$$\int_0^1 \frac{\ln(1-t)}{t^2\sqrt{t}}$$
$\boxed{\text{diverges}}$. At $t=0$, $\ln(1-t)\approx -t$, so the integrand is $t^{-\frac{3}{2}} < -1$ so it diverges.

8.
$$\int_0^1 \frac{t+\ln(1-t)}{t^2\sqrt{t}} = -\sum_{n=2}^{\infty}\frac{1}{n}\cdot\frac{2}{2n-3} = -\sum_{n=2}^{\infty}\left(\frac{2}{3n}-\frac{4}{3(2n-3)}\right) = \frac{-2}{3}-\frac{4\ln 2}{3} = \boxed{-\frac{2(1+2\ln 2)}{3}}$$